Assignment No 5

Title: Crop Yield Prediction using Machine Learning Models

Aim

To predict **crop yield (tonnes/hectare)** based on **rainfall** and **fertilizer usage** using different regression models:

- Linear Regression
- Ridge Regression
- Random Forest Regressor

using the Crop Recommendation dataset.

Objective

- To understand regression techniques in Machine Learning.
- To compare the performance of Linear Regression, Ridge Regression, and Random Forest Regressor.
- To evaluate models using performance metrics such as:
    - Mean Absolute Error (MAE)
    - Mean Squared Error (MSE)
    - R^2 Score

Prerequisites

- Python 3.x
- Jupyter Notebook / Google Colab
- Libraries:
    - numpy
    - pandas
    - matplotlib

- seaborn
- scikit-learn

Install required libraries:

pip install numpy pandas matplotlib seaborn scikit-learn

Theory

Linear Regression

Linear Regression establishes a linear relationship between input features and output:

$$Y = b0 + b1X1 + b2X2$$

Where:

- Y = Crop Yield
- X1 = Rainfall
- X2 = Fertilizer Usage

Ridge Regression

Ridge Regression is a regularized version of Linear Regression.

It adds a penalty term to reduce overfitting:

$$Loss = RSS + \lambda \sum b^2$$

It helps when features are correlated.

Random Forest Regressor

Random Forest is an ensemble learning method that:

- Builds multiple decision trees
- Combines their predictions
- Reduces overfitting

- Works well for nonlinear relationships

Algorithm

1. Import required libraries
2. Load the dataset
3. Select features:
    - Rainfall
    - Fertilizer
4. Select target variable:
    - Yield
5. Split dataset into training and testing sets
6. Train models:
    - Linear Regression
    - Ridge Regression
    - Random Forest Regressor
7. Predict on test data
8. Evaluate models
9. Compare results

Performance Metrics

Metric	Description
MAE	Average absolute error
MSE	Average squared error
R^2 Score	Model accuracy (closer to 1 is better)

Expected Result

- Linear Regression performs well for linear relationships.
- Ridge Regression reduces overfitting.
- Random Forest Regressor gives better accuracy for nonlinear patterns.
- Random Forest generally provides the best R^2 score.

Conclusion

In this experiment, we predicted crop yield using three regression models.

- Linear Regression is simple and interpretable.
- Ridge Regression handles multicollinearity.
- Random Forest gives better performance for complex data.

Thus, Random Forest Regressor is typically the best model for crop yield prediction when the relationship is nonlinear.